

ICS 4111: Embedded Systems & IoT

Practical Exercise 2 – Circuit Design

Part A – Circuit Calculations

Circuit (a) – Series Circuit

Given: $V = 125\text{ V}$, $R1 = 20\ \Omega$, $R2 = 30\ \Omega$, $R3 = 50\ \Omega$ (all in series)

1. Total resistance

$$R_T = R1 + R2 + R3 = 20 + 30 + 50 = 100\ \Omega$$

2. Total current from the supply

$$I_T = V / R_T = 125 / 100 = 1.25\text{ A}$$

3. Current through each resistor

In a series circuit the current is the same at every point, so:

$$I1 = I2 = I3 = I_T = 1.25\text{ A}$$

4. Voltage drop across each resistor

$$V1 = I1 \times R1 = 1.25 \times 20 = 25\text{ V}$$

$$V2 = I2 \times R2 = 1.25 \times 30 = 37.5\text{ V}$$

$$V3 = I3 \times R3 = 1.25 \times 50 = 62.5\text{ V}$$

5. Power dissipated by each resistor

$$P1 = V1 \times I1 = 25 \times 1.25 = 31.25\text{ W}$$

$$P2 = V2 \times I2 = 37.5 \times 1.25 = 46.875\text{ W}$$

$$P3 = V3 \times I3 = 62.5 \times 1.25 = 78.125\text{ W}$$

Resistor	Resistance (Ω)	Current (A)	Voltage Drop (V)	Power (W)
R1	20	1.25	25	31.25
R2	30	1.25	37.5	46.875
R3	50	1.25	62.5	78.125

Circuit (b) – Parallel Circuit

Given: $V = 125\text{ V}$, $R1 = 20\ \Omega$, $R2 = 100\ \Omega$, $R3 = 50\ \Omega$ (all in parallel)

1. Total resistance

$$1/R_T = 1/R1 + 1/R2 + 1/R3 = 1/20 + 1/100 + 1/50$$

$$1/R_T = 0.05 + 0.01 + 0.02 = 0.08$$

$$R_T = 1 / 0.08 = 12.5\ \Omega$$

2. Total current from the supply

$$I_T = V / R_T = 125 / 12.5 = 10\text{ A}$$

3. Current through each resistor

In a parallel circuit each branch has the full supply voltage across it, so:

$$I1 = V / R1 = 125 / 20 = 6.25\text{ A}$$

$$I2 = V / R2 = 125 / 100 = 1.25\text{ A}$$

$$I3 = V / R3 = 125 / 50 = 2.5\text{ A}$$

4. Voltage drop across each resistor

Each branch is connected directly across the supply, so:

$$V_1 = V_2 = V_3 = 125 \text{ V}$$

5. Power dissipated by each resistor

$$P_1 = V_1 \times I_1 = 125 \times 6.25 = 781.25 \text{ W}$$

$$P_2 = V_2 \times I_2 = 125 \times 1.25 = 156.25 \text{ W}$$

$$P_3 = V_3 \times I_3 = 125 \times 2.5 = 312.5 \text{ W}$$

Resistor	Resistance (Ω)	Current (A)	Voltage Drop (V)	Power (W)
R1	20	6.25	125	781.25
R2	100	1.25	125	156.25
R3	50	2.5	125	312.5

Part B – Schematic Diagrams

1. Series Circuit (a) Schematic

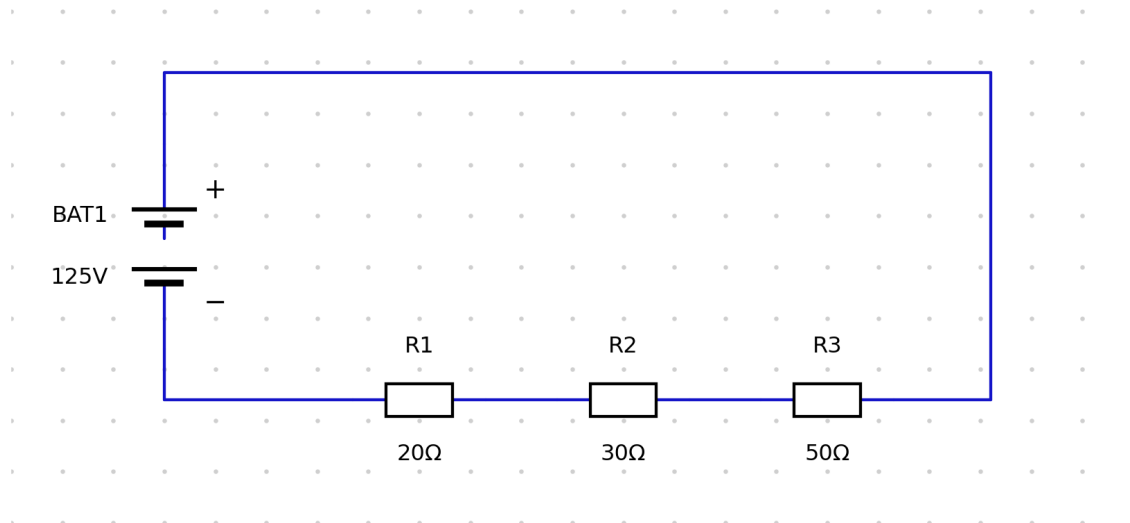


Figure 1: Series circuit – 125 V source with R1, R2, R3 in series.

Ref	Component	Value / Type	Notes
BAT1	DC Voltage Source	125 V	Main supply
R1	Resistor	20 Ω	Series, 1.25 A
R2	Resistor	30 Ω	Series, 1.25 A
R3	Resistor	50 Ω	Series, 1.25 A

2. Parallel Circuit (b) Schematic

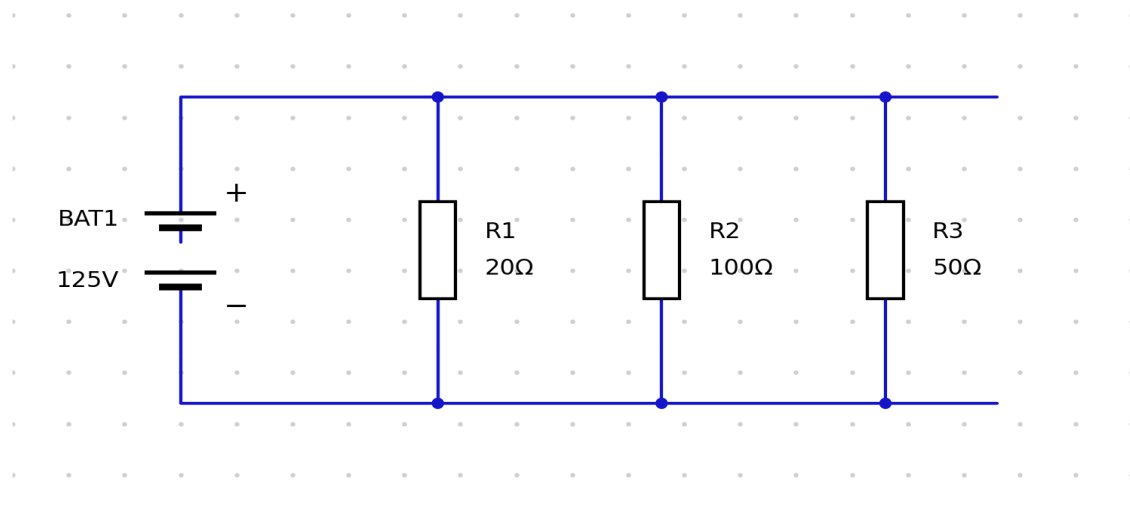
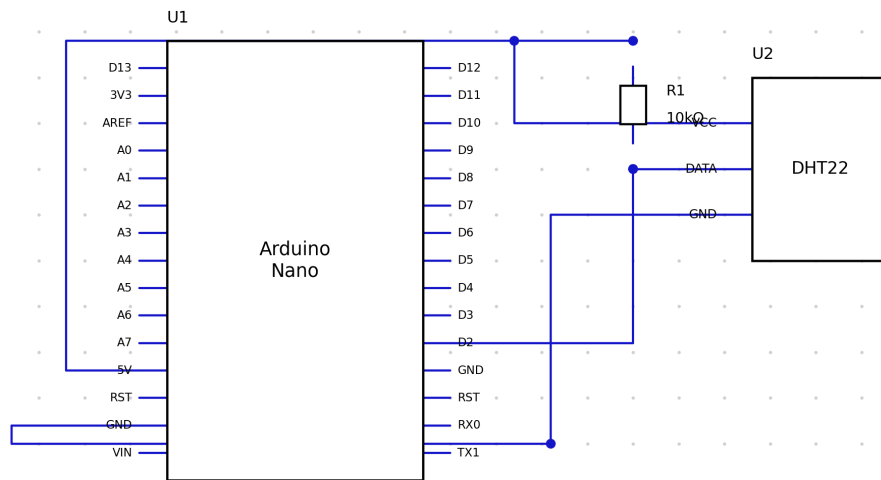


Figure 2: Parallel circuit – 125 V source with R1, R2, R3 in parallel.

Ref	Component	Value / Type	Notes
BAT1	DC Voltage Source	125 V	Main supply
R1	Resistor	20 Ω	Parallel branch, 6.25 A
R2	Resistor	100 Ω	Parallel branch, 1.25 A
R3	Resistor	50 Ω	Parallel branch, 2.5 A

3. Arduino Nano – DHT22 Interface Schematic



Note: DHT22 requires 3.3V-5V supply. Powered here from Nano 5V pin.

Figure 3: Arduino Nano connected to a DHT22 sensor, with a 10 kΩ pull-up resistor on the data line.

Ref	Component	Value / Type	Notes
U1	Arduino Nano	ATmega328P board	Microcontroller
U2	DHT22 (AM2302)	Temp/Humidity sensor	Supply range 3.3–5 V
R1	Resistor	10 kΩ	Pull-up on DATA line